

Graph-Based Adaptive Fusion of GNSS and VIO Under Intermittent GNSS-Degraded Environment

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Abstract—Consistent and accurate global positioning is a crucial problem for autonomous vehicles and robots. It is especially challenging in situations with a global navigation satellite system (GNSS) being intermittently degraded. We propose an adaptive fusion system, namely, GNSS/visual-inertial navigation system (GVINS), which adaptively fuses GNSS and visual-inertial odometry (VIO) to achieve consistent and accurate global positioning, even in GNSS intermittently degraded scenarios. Compared with existing methods, GVINS can provide positioning under an Earth-fixed geographic coordinate, rather than a local tangent plane (LTP) coordinate, as long as any GNSS measurement presents in the trajectory. To adaptively fuse VIO and GNSS, we first use an inertial measurement unit (IMU) preintegration-based depth uncertainty estimation method to evaluate the accuracy of VIO. Then, in the optimization backend, we perform an innovative overparameterized, 15 degree-of-freedom pose-graph fusion. An alternating minimization (AM) algorithm is used to efficiently solve this problem. Evaluation results on both public and custom-built data sets demonstrate that GVINS outperforms state-of-the-art fusion methods in both accuracy and stability in GNSS intermittently degraded environments.

Index Terms—Anti-interference, global navigation satellite system (GNSS), pose-graph optimization, sensor fusion, visual-inertial odometry (VIO).

I. INTRODUCTION

IN THE past decade, using visual odometry (VO), visual-inertial odometry (VIO), and vision-based simultaneous localization and mapping (V-SLAM) systems under global navigation satellite system (GNSS)-denied environments have been extensively studied, which have produced fruitful outcomes, such as ORB-SLAM [1], [2], OKVIS [3], IVPR [4], and VINS-mono [5]. Such systems have become basic components in autonomous vehicles and robots to locate themselves under complex environments.

As the local VO/VIO/V-SLAM methods commonly have drawbacks like drifting, nondrifting global positioning methods can help to overcome the drifting problem and improve

both the accuracy and robustness of localization. Accordingly, to take advantage of both local and global methods, many fusion methods have been proposed [6]–[8].

As local methods output positioning under local frame only, applications involving global coordinates would face predicaments when using local methods alone. In particular, local methods reset the frame origin every time the starting point changes and consequently reform the pose estimation even if the vehicle stays at the same point within the same scene. To provide consistent and authentic localization under global coordinates, an effective way is to resort to fusion systems that can fuse local pose estimation and global pose information. Such a fusion system is expected to take advantage of the local estimation module and, in the meantime, can incorporate information from global sensors to eliminate drift and achieve global localization.

Though much progress has been made in the studies of such fusion systems, there still exist some limitations in existing works. One is the consideration of intermittent GNSS-denied scenario. In practical applications, autonomous vehicles need to travel through diverse environments, including but not limited to indoors, pathways, highways, and urban canyons. Unlike the ON-and-NON-OFF GNSS outage assumption used in existing works [7], [9]–[11], intermittent GNSS degradation is closer to the practical circumstance, where the quality of GNSS measurements varies dramatically. To fuse GNSS into a vision-based localization system, threshold-based rejection and switching [10] is the most commonly used method, yet contributes little to the accuracy improvement. Such a method drops all unqualified measurements, which would decrease the usability of the system.

Moreover, there is another difficulty for a VO/VIO/VSLAM system to be adaptively fused with GNSS measurements. That is to provide an efficient and explicit estimation of local pose uncertainty. Though this estimation is theoretically feasible, most implementations in practice fix the mapping uncertainty for computational efficiency, which leads to overconfidence in pose uncertainty [12]. Besides, the divergence of image processing methods used in different VO/VIO/V-SLAM systems hinders unified modeling of spatial uncertainty.

To address these problems, we proposed a fusion system for global positioning, namely, GNSS-visual-inertial navigation system (GVINS), which uses nonlinear graph-based

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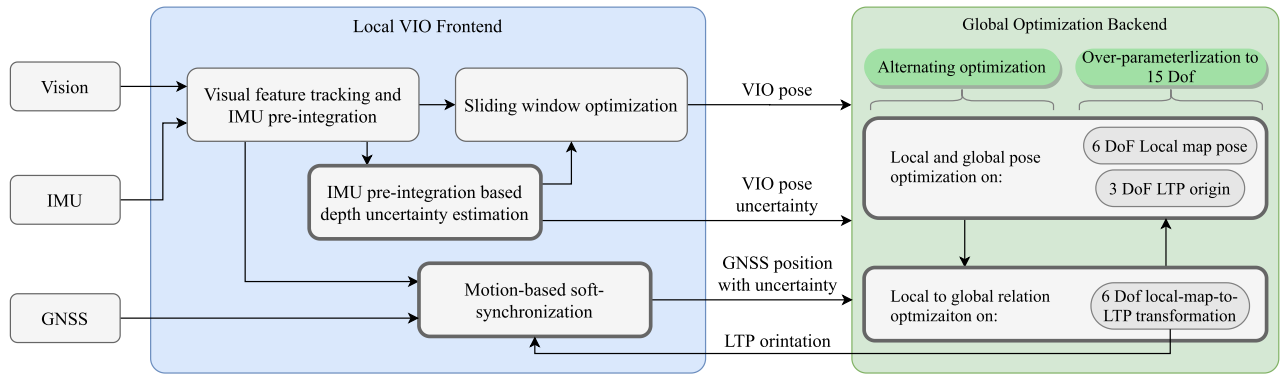


Fig. 1. System block diagram illustrates all the modules introduced in this article. Modules with bold border are highlighted innovations in this work.

optimization to fuse GNSS and VIO. Compared with filter-based fusion methods, such as [12]–[14], the optimization-based methods can provide higher accuracy, but compromise on computational efficiency. This tradeoff has been well demonstrated in [15]. However, since the optimization scheme is based on batch estimation, it benefits robustness under the concerned intermittent GNSS-degraded environment.

The proposed system contains a VIO frontend and an adaptive fusion backend. The VIO frontend uses inertial measurement unit (IMU) motion to provide a computationally efficient estimation of pose uncertainty via inverse-depth uncertainty analysis. This VIO frontend is also in charge of synchronizing the vision and GNSS measurements. For the batch approaches, such as [2] and [5], the uncertainty propagation is usually neglected since the uncertainty grows unboundedly and becomes numerically problematic as time goes on. However, the absence of this uncertainty would degrade the following fusion systems, such as [7], especially in complex environments. Meanwhile, other approaches, such as [3], propagate the uncertainty relatively but raise the computation consumption. Therefore, as an eclectic choice, our approach estimates the uncertainty using ego-motion characteristics rather than propagates it through frames.

Then, on the fusion backend, we first formulated a 15 degree-of-freedom (DoF) optimization state, which is specialized for intermittent GNSS-degraded environment. Besides the classical 6 DoF local pose, this 15 DoF state includes an overparameterized 6 DoF local-to-global transformation and an extra 3 DoF of global origin bias. The former one of the extra DoF improves the robustness of the system, while the latter improves global accuracy and computation efficiency. Compared with existing methods, such as [7] and [8], the proposed formulation relaxes the assumption that the system must be initialized under a decent GNSS environment, which makes the system more applicable to intermittent GNSS-degraded environment. Note that the proposed system needs GNSS measurement presenting (though degraded) in the trajectory to obtain the global position result; otherwise, only a local result can be calculated.

Finally, we employed a dual-stage alternating minimization (AM) procedure to efficiently optimize the global problem by splitting the coupled variables.

In summary, the contributions of this work are highlighted as follows.

- 1) An IMU pre-integration-based relative uncertainty estimation method. It estimates key-point depth uncertainty independently of IMU pre-integration terms and propagates it into relative pose uncertainty. It gives efficient and motion-only uncertainty estimation.
- 2) An overparameterized model of 15 DoF on pose estimation, including an extra 6 DoF on the local-map-to-local tangent plane (LTP) pose and a 3 DoF on LTP origin bias. This overparameterization makes it possible to initialize the system under the GNSS-degraded environment while reducing the computational cost.
- 3) A dual-stage alternating optimization algorithm to efficiently estimate global pose under Earth-fixed frame.
- 4) An open-source tool¹ for interpolating the KITTI data set [16]. This tool can also generate artificial noise to GNSS trajectories in given areas.

The rest of this article was organized as follows. In Section II, we presented an overview of the system. In Section III, we first reviewed a typical IMU pre-integration-based, sliding window VIO, then introduced the soft-synchronization mechanism. In Section IV, we presented IMU pre-integration-based relative uncertainty estimation. In Section V, we explained the overparameterized formulation and discussed it with others. Later, the dual-stage alternating optimization was formulated in Section VI. Implementation details and experimental results were provided in Section VII. Finally, in Section VIII, we concluded this article with a brief discussion and possible future research directions. In addition, the algorithm used in the KITTI processing tool was introduced in the Appendix.

II. SYSTEM OVERVIEW

The overall structure of the proposed system is shown in Fig. 1. This system contains two loosely-coupled modules, namely, the VIO frontend and the optimization backend. The frontend of the system takes vision and six-axis IMU (three-axis accelerometer and three-axis gyroscope) as local observation input and GNSS as global observation input. The frontend processing begins with an IMU pre-integration-based sliding-window VIO module (Section III). It first fuses IMU and camera data and tightly couples them to give VIO pose estimation. In consideration of cost efficiency, we took stamped, instead

¹<https://github.com/gauxonz/KITTI-interp>

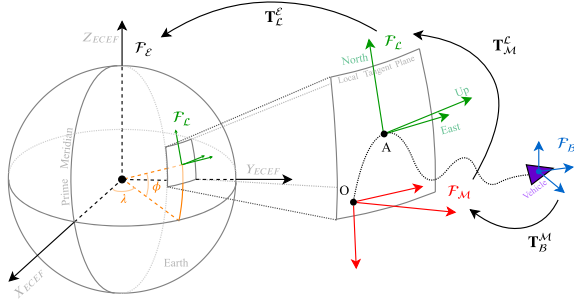


Fig. 2. Coordinates involved in vehicle motion with the translation matrices between each other.

of triggered, hard-synchronized sensor data as the input of our system. Therefore, a software synchronization mechanism (Section III-B) was adopted in the frontend to synchronize the input GNSS data to the local VIO pose estimation. Moreover, this frontend makes estimation not only on ego-motion but on localization uncertainty as well. In the uncertainty estimation (Section IV), the IMU pre-integration method [17] was used to estimate visual inverse-depth uncertainty. Then, this uncertainty was further propagated to the backend for adaptive fusion.

On the backend side, the optimization takes two pairs of inputs, the local pose estimation with its uncertainty under the local vision frame and the GNSS position measurement synchronized with the VIO output. We proposed an over-parameterized formulation (Section V) on the states to be optimized. Then, to avoid the gradient diffusion caused by the high DoF, we adopted a dual-stage strategy (Section VI) to facilitate the optimization system to descend in the correct direction.

A. Notations and Pose Representations

We represented rotations from frame A to frame B by either rotation matrix $\mathbf{R}_A^B$ or $\mathbf{q}_A^B$. The conversion between these two forms, $\mathbf{q}_A^B = \mathbf{q}_A^B\{\mathbf{R}_A^B\}$ and $\mathbf{R}_A^B = \mathbf{R}_A^B\{\mathbf{q}_A^B\}$, has been well-summarized in [18]. This conversion happens by default, and thus, we abbreviated the conversion operator $q\{\cdot\}$ and $R\{\cdot\}$ if there was no ambiguity. Moreover, when a 4×4 homogeneous translation $\mathbf{T}$ acts on a 3×1 pose $\mathbf{p}$, the homogeneous conversion also happens implicitly if not specifically mentioned.

B. Frames Used in Local and Global Positioning

We first clarified all coordinate frames involved in this work. The main idea of this part was first introduced in our previous work [8], we modified and reviewed it here for completeness.

As shown in Fig. 2, a vehicle (purple triangular) has a body frame $\mathcal{F}_B$, which is an inertial Cartesian frame. $\mathcal{F}_B$ can be defined arbitrarily. All the other sensors on the vehicle can be calibrated into $\mathcal{F}_B$ assuming rigidly connected through methods, such as [19] and [20]. When the vehicle is initialized and sets off from point O , a local map frame $\mathcal{F}_M$ is formed. $\mathcal{F}_M$ is usually initialized by the visual system and contains little or incomplete information about geographic direction.

If an accelerometer exists, the “down” direction can be estimated; meanwhile, the orientation cannot be informed unless a compass exists. $\mathcal{F}_B$ and $\mathcal{F}_M$ should be identical at the initialization when ignoring inner sensor calibration errors.

Then, at another point A , when the global measurement (such as GNSS) is first received, an LTP frame $\mathcal{F}_L$ like East–North–up or North–East–down is registered. There are two points worth noticing on $\mathcal{F}_L$. First, although many works consider $\mathcal{F}_L$ as a “global frame,” it is essentially a local frame since all the translations are related to the initial point A . Second, considering the common situation of “initialized indoors and going outdoors,” point A and O are not necessarily being identical. This is a complement to our previous work [8].

In the end, the Earth-referenced frame $\mathcal{F}_E$ is considered as the global frame. Again, it can have multiple configurations from the Cartesian Earth-centered–Earth-fixed (ECEF) frame to the horizontal longitude–latitude–altitude (LLA) frame, which can be easily converted into each other.

Thus, an integrity transformation of a point $\mathbf{p}^B = [p_x \ p_y \ p_z]^T$ from the body frame to the global Earth frame $\mathcal{F}_E$ is

$$\begin{aligned} \tilde{\mathbf{p}}^E &= \mathbf{T}_L^E \mathbf{T}_M^L \mathbf{T}_B^M \tilde{\mathbf{p}}^B \\ &= \begin{bmatrix} \mathbf{R}_L^E & \mathbf{t}_L^E \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}_M^L & \mathbf{t}_M^L \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{R}_B^M & \mathbf{t}_B^M \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \mathbf{p}^B \\ 1 \end{bmatrix} \quad (1) \end{aligned}$$

where $\tilde{(\cdot)}$ is the homogeneous representation of points, and $\mathbf{T}_L^E$, $\mathbf{T}_M^L$, and $\mathbf{T}_B^M$ are the homogeneous translation matrices between frames, shown as black solid symbols in Fig. 2. $\mathbf{R}_B^M$ and $\mathbf{R}_M^L$ are 3×3 rotation matrices with 3 DoF and $\mathbf{t}_B^M$, $\mathbf{t}_M^L$ are 3×1 translation vectors with 3 DoF. Usually, a global measurement is given under the curvilinear coordinate longitude–latitude–height (LLH) as $\mathbf{p} \triangleq [\lambda \ \phi \ h]^T$ (orange symbols shown in Fig. 2), and the metric representations $\mathbf{t}_L^E$ can be transformed by $\mathbf{p}$ as $\mathbf{t}_L^E = C(\mathbf{p})$. Then, $\mathbf{R}_L^E$ means the rotation transform from the local ground world frame $\mathcal{F}_L$ to the ECEF world frame $\mathcal{F}_E$, which is not independent on $\mathbf{p}$. This bonding is represented as $\mathbf{R}_L^E = M(\mathbf{p})$. Therefore, $\mathbf{R}_L^E$ and $\mathbf{t}_L^E$ together only have 3 DoF in total, determined by $\mathbf{p}$. These two Cartesian and polar coordinates transformation function $C(\cdot)$ and $M(\cdot)$ are derived in [21].

From (1), we can summarize that, the local estimation is given under $\mathcal{F}_M$ as $\mathbf{T}_B^M$, which would be discussed in Section III. Meanwhile the global optimization, which involves the GNSS measurement under $\mathcal{F}_E$ as $\mathbf{t}_B^E$, would be discussed in Section V.

III. LOCAL VISUAL-INERTIAL ODOMETRY AND SENSOR SOFT-SYNCHRONIZATION

A. Local VIO Formulation

The vision and IMU data were first coupled by a VIO to produce local pose estimation. We adopted an existing state-of-the-art VIO algorithm, VINS-Mono [5], as a development foundation. This part contains no innovation in this work. However, for completeness, we briefly introduced it in this section.

The VIO estimates the vehicle’s poses on visual keyframes and visual features’ depth within a sliding window. Assuming

all the sensors are well-calibrated into IMU, the states are defined as

$$\mathcal{X}_{\mathcal{M}} = [\mathbf{x}_0^T, \mathbf{x}_1^T, \dots, \mathbf{x}_n^T, \lambda_0, \lambda_1, \dots, \lambda_m]^T$$

$$\mathbf{x}_k = [\mathbf{t}_{B_k}^T, \mathbf{v}_{B_k}^T, \mathbf{q}_{B_k}^T, \mathbf{b}_a^T, \mathbf{b}_g^T]^T, \quad k \in [0, n] \quad (2)$$

where $\mathcal{X}_{\mathcal{M}}$ represents the states set under $\mathcal{F}_{\mathcal{M}}$. The k th state $\mathbf{x}_k$ consists of a translation $\mathbf{t}_{B_k}^M$, a velocity $\mathbf{v}_{B_k}^M$, and an orientation $\mathbf{q}_{B_k}^M$ from $\mathcal{F}_{B_k}$ to $\mathcal{F}_{\mathcal{M}}$. It should be noticed that we used quaternion in states to represent the orientation for a condensed parameterization, as mentioned in Section II-A. Therefore, $\mathbf{q}_{B_k}^M$ here is directly converted from $\mathbf{R}_{B_k}^M$ as in (1). $\mathbf{b}_a$ and $\mathbf{b}_g$ are accelerometer bias and gyroscope bias, respectively, and λ is the inverse depth of visual features.

As all the motions mentioned here are from $\mathcal{F}_B$ to $\mathcal{F}_{\mathcal{M}}$, the coordinate denotation $(\cdot)_{\mathcal{B}}^M$ would be omitted from now for concision.

The propagation from the k th frame to the $(k+1)$ th frame has the form of

$$\begin{bmatrix} \mathbf{t}_{k+1} \\ \mathbf{v}_{k+1} \\ \mathbf{q}_{k+1} \\ \mathbf{b}_{a_{k+1}} \\ \mathbf{b}_{g_{k+1}} \end{bmatrix} = \begin{bmatrix} \mathbf{t}_k + \mathbf{v}_k \Delta t - \frac{1}{2} \mathbf{g} \Delta t^2 + \mathbf{q}_k \boldsymbol{\alpha}_k^{k+1} \\ \mathbf{v}_k + \mathbf{g} \Delta t + \mathbf{q}_k \boldsymbol{\beta}_k^{k+1} \\ \mathbf{q}_k \mathbf{q}_k^{k+1} \\ \mathbf{b}_{a_k} \\ \mathbf{b}_{g_k} \end{bmatrix} \quad (3)$$

where $\mathbf{g}$ is the gravity, and Δt is the time interval between neighbor visual keyframes. $\boldsymbol{\alpha}_k^{k+1}$, $\boldsymbol{\beta}_k^{k+1}$, and $\mathbf{q}_k^{k+1}$ are the pre-integration components, given by

$$\boldsymbol{\alpha}_k^{k+1} = \iint_{t \in [k, k+1]} (\mathbf{q}_k^t \mathbf{a}_t) \delta t^2$$

$$\boldsymbol{\beta}_k^{k+1} = \int_{t \in [k, k+1]} (\mathbf{q}_k^t \mathbf{a}_t) \delta t$$

$$\mathbf{q}_{b_i b_j} = \int_{t \in [k, k+1]} \mathbf{q}_k^t \otimes \left[\frac{0}{\frac{1}{2} \boldsymbol{\omega}_t} \right] \delta t \quad (4)$$

where $\mathbf{a}_t$ and $\boldsymbol{\omega}_t$ are body acceleration and angular velocity at time t , respectively, and δt is IMU time interval between IMU frames. $\otimes$ represents the quaternion product.

The local estimation is formulated as the following nonlinear problem:

$$\min_{\mathcal{X}_{\mathcal{M}}} \left\{ \underbrace{\rho(\|\mathbf{r}_p - \mathbf{H}_p \mathcal{X}_{\mathcal{M}}\|_{\Sigma_p}^2)}_{\text{prior error}} + \underbrace{\sum_{k \in \mathcal{I}} \rho(\|\mathbf{r}_{\mathcal{I}}(\hat{\mathbf{z}}_{B_k B_{k+1}}, \mathcal{X}_{\mathcal{M}})\|_{\Sigma_{B_k B_{k+1}}}^2)}_{\text{IMU error}} + \underbrace{\sum_{(l,j) \in \mathcal{C}} \rho(\|\mathbf{r}_{\mathcal{C}}(\hat{\mathbf{z}}_l^{c_j}, \mathcal{X}_{\mathcal{M}})\|_{\Sigma_l^{c_j}}^2)}_{\text{camera error}} \right\} \quad (5)$$

which contains three parts of residuals, including the prior part, the inertial part, and the visual part. $\rho(\cdot)$ stands for a robust

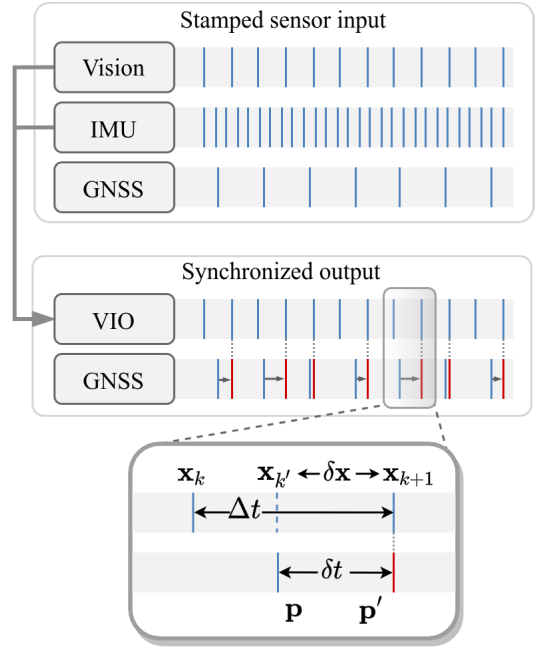


Fig. 3. Synchronization strategy used in the frontend. Blue ticks are original data epochs, while red ones are synchronized.

loss function [22]. $\|\cdot\|_{\Sigma}^2$ represents the squared Mahalanobis distance with covariance matrix Σ . Besides, $r_{\mathcal{I}}(\hat{\mathbf{z}}_{B_k B_{k+1}}, \mathcal{X}_{\mathcal{M}})$ and $\mathbf{r}_{\mathcal{C}}(\hat{\mathbf{z}}_l^{c_j}, \mathcal{X}_{\mathcal{M}})$ are local residuals of IMU and visual measurements $\hat{\mathbf{z}}_{B_k B_{k+1}}$, $\hat{\mathbf{z}}_l^{c_j}$, respectively. $\{\mathbf{r}_p, \mathbf{H}_p\}$ is the prior information from marginalization. A detailed explanation can be found in [5].

B. Soft Synchronization of VIO and GNSS

As mentioned in Section II, we took stamped sensor inputs considering the hardware cost. Therefore, data synchronization was required to ensure accuracy. In general, typical cameras have a frame rate of 30 Hz, IMU has a typical output rate of 100 Hz, while a GNSS receiver typically works at 1 Hz. Due to the high frequency of IMU, the synchronization between visual sensors and IMU inside the VIO system is usually fulfilled by a midpoint interpolation, as described in [5]. Then, to couple the VIO ego-motion estimation with GNSS data, we further considered their synchronization. Existing methods, such as [7], [8], and [23], use a simple threshold-based tolerance to reject all the measurements with a certain time-difference. These methods work well when the vehicle's speed is relatively slow compared with the error of GNSS measurement or when the GNSS receiver has a high refresh rate compared with the VIO output rate. Both of the assumptions limit the application in general situations.

Therefore, the proposed system uses a linear interpolation strategy to synchronize the GNSS and VIO outputs, as shown in Fig. 3. $\mathbf{x}_k$ represents the VIO output states at frame k as defined in (2). Δt and δt are the time intervals of VIO estimation and GNSS measurement, respectively. When a GNSS LLH measurement $\mathbf{p}$ comes, we do not pass it to the backend immediately but wait until the closest next VIO frame

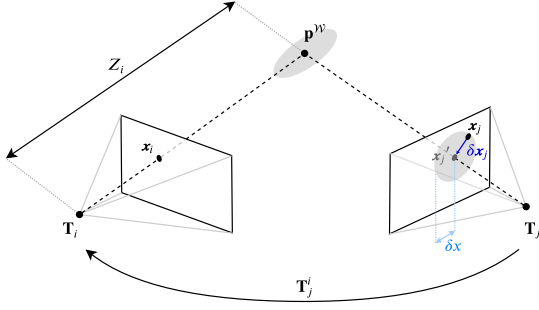


Fig. 4. Two-view geometry with matching uncertainty.

$k + 1$ arrive. Then, the increment state $\delta \mathbf{x}$ from the linearly interpolated VIO states $\mathbf{x}_{k'}$ at $\mathbf{p}$'s timestamp to $\mathbf{x}_{k+1}$ at $\mathbf{p}'$'s timestamp is given by

$$\delta \mathbf{x} = \frac{\delta t}{\Delta t} (\mathbf{x}_{k+1} \boxminus \mathbf{x}_k) \quad (6)$$

where $\boxminus$ denote the increase and decrease operators for position, velocity, and quaternion rotation. Next, an LTP coordinate is initialized at point $\mathbf{p}$, with $\mathbf{t}_L^E = C(\mathbf{p})$ and $\mathbf{R}_L^E = M(\mathbf{p})$. Then, the predicted position $\mathbf{p}'$ is given as

$$\mathbf{p}' = P(\mathbf{R}_L^E \mathbf{R}_{\mathcal{M}}^L \delta \mathbf{t} + \mathbf{t}_L^E) \quad (7)$$

where $P(\cdot) = C^{-1}(\cdot)$ is the conversion from Cartesian to polar geodetic coordinates as introduced in [21]. $\mathbf{R}_{\mathcal{M}}^L$, the LTP coordinate orientation is the optimization result from the global fusion (Section VI), as shown in Fig. 1. Once $\mathbf{p}'$ is calculated, we republish it with a synchronized timestamp with $\mathbf{x}_{k+1}$.

IV. IMU PRE-INTEGRATION-BASED RELATIVE UNCERTAINTY ESTIMATION

It is well known that the accuracy of VO/VIO/V-SLAM-based localization is influenced by the vehicle's motion pattern, especially the twist-velocity on the yaw axis [24], [25]. Depth estimation of feature points generally gets numerically unstable when the baseline is short, e.g., during pure rotational motion. The uncertainty of the pose estimation will rise at the same time. In this section, we analyzed this influence and proposed an IMU pre-integration-based uncertainty estimation method.

As shown in Fig. 4, there is a landmark l with position $\mathbf{p}^w = [X \ Y \ Z]^T$ under (local) world frame $\mathcal{F}_w$. At this moment, all states about frame i have already been solved, and the system is about to estimate frame j . Let $\mathbf{R}_i^w$ and $\mathbf{t}_i^w$ represent the rotation and translation at frame i of the vehicle under $\mathcal{F}_w$. Then, we obtain

$$\mathbf{p}^w = \mathbf{R}_i^w \mathbf{K}^{-1} Z_i \tilde{\mathbf{p}}_i + \mathbf{t}_i^w = \mathbf{R}_j^w \mathbf{K}^{-1} Z_j \tilde{\mathbf{p}}_j + \mathbf{t}_j^w \quad (8)$$

where Z_i is the depth of $\mathbf{p}^w$ at i th keyframe. $\tilde{\mathbf{p}}_i = [u_i \ v_i \ 1]^T$ is the homogeneous representation of a feature point on the image plane. $\mathbf{K}$ is the camera's intrinsics. Taking $\mathbf{x}_i = \mathbf{K}^{-1} \tilde{\mathbf{p}}_i$, we obtain

$$Z_i \mathbf{x}_i = \mathbf{R}_j^i Z_j \mathbf{x}_j + \mathbf{t}_j^i. \quad (9)$$

By multiplying $[\mathbf{x}_i]_{\times}$ on both sides of (9), we have

$$Z_j [\mathbf{x}_i]_{\times} \mathbf{R}_j^i \mathbf{x}_j = -[\mathbf{x}_i]_{\times} \mathbf{t}_j^i \quad (10)$$

where $[\mathbf{x}]_{\times}$ represents the skew-symmetric matrix of $\mathbf{x}$. Therefore, the inverse-depth at the j th frame d_j satisfies

$$-[\mathbf{x}_i]_{\times} \mathbf{t}_j^i d_j = [\mathbf{x}_i]_{\times} \mathbf{R}_j^i \mathbf{x}_j. \quad (11)$$

Here, unlike the triangulation, we are not trying to optimize $\mathbf{R}_j^i$ and $\mathbf{t}_j^i$, but to estimate the uncertainty on d_j using the $\mathbf{R}_j^i$ and $\mathbf{t}_j^i$ from IMU integration. Assuming the uncertainty on feature matching has a constant value of δp pixels, then on the normalized plane, the uncertainty forms a circle with radius $\delta x = f^{-1} \delta p$, where f is the camera focal length. This is illustrated as the gray disk on camera plane j in Fig. 4. Consequentially, the uncertainty on inverse depth δd satisfies

$$-[\mathbf{x}_i]_{\times} \mathbf{t}_j^i \delta d = [\mathbf{x}_i]_{\times} \mathbf{R}_j^i \delta \mathbf{x}_j \quad (12)$$

where $\delta \mathbf{x}_j$ is the matching error vector on frame j , as shown in Fig. 4 with a dark blue arrow. It should be noticed that $\delta \mathbf{x}_j$ is a 3×1 vector but with the last element being 0. Therefore $\|\delta \mathbf{x}_j\| = \|\delta \mathbf{x}_{j(1:2)}\| \leq \delta x$. By the property of cross product, we have

$$\begin{aligned} \delta d &= \frac{\|[\mathbf{x}_i]_{\times} \mathbf{R}_j^i \delta \mathbf{x}_j\|}{\|[\mathbf{x}_i]_{\times} \mathbf{t}_j^i\|} \\ &= \frac{\|\mathbf{x}_i\| \|\mathbf{R}_j^i \delta \mathbf{x}_j\| |\sin \theta|}{\|[\mathbf{x}_i]_{\times} \mathbf{t}_j^i\|} \end{aligned} \quad (13)$$

where θ is the angle between $\mathbf{x}_i$ and $\mathbf{R}_j^i \delta \mathbf{x}_j$. The numerator of (13) reaches its maximum value when $\mathbf{R}_j^i \delta \mathbf{x}_j$ is perpendicular to $\mathbf{x}_i$, aka. $|\sin \theta| = 1$. Thus

$$\delta d \leq \frac{\|\mathbf{x}_i\| \|\mathbf{R}_j^i \delta \mathbf{x}_j\|}{\|[\mathbf{x}_i]_{\times} \mathbf{t}_j^i\|}. \quad (14)$$

As $\|\delta \mathbf{x}_j\| \leq \delta x$ is given by assumption, and $\mathbf{R} \mathbf{R}^T = \mathbf{I}$ for any rotation matrix $\mathbf{R}$, we have

$$\delta d \leq \frac{\|\mathbf{x}_i\|}{\|[\mathbf{x}_i]_{\times} \mathbf{t}_j^i\|} \cdot \delta x. \quad (15)$$

All the matrix norms earlier are the Hilbert–Schmidt norm.

From (15), we can draw a qualitative conclusion that the interframe inverse-depth uncertainty of a tracked feature point is inversely proportional to the cross product of the normalized direction under the camera frame and the vehicle's motion vector and proportional to the feature matching uncertainty δx . The feature matching uncertainty varies from different feature extraction methods [26], [27]; thus, in practice, it is usually assigned with a constant empirical value. In this article, we concentrate on the motion-related part only. $\mathbf{t}_j^i$ in (15) is no other than the translation between frames, and therefore, can be represented by IMU pre-integration terms, through (3) and (4), as

$$\delta d = \frac{\|\mathbf{x}_i\|}{\|[\mathbf{x}_i]_{\times} (\mathbf{v}_i^w \Delta t - \frac{1}{2} \mathbf{g} \Delta t^2 + \mathbf{q}_i \boldsymbol{\alpha}_i^j)\|} \delta x. \quad (16)$$

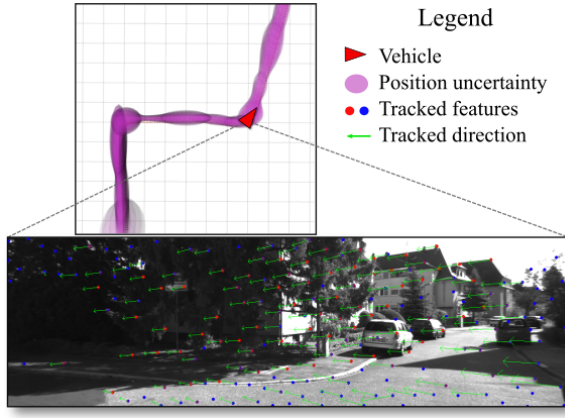


Fig. 5. Runtime screenshot demonstrating the calculated uncertainty increment during turning.

As δd represents the uncertainty of landmark l from the perspective of j th frame, it can be directly used in the covariance of visual residual term in (5) with a proper weight

$$\Sigma_j^c = W_c \delta d_l^j \mathbf{1}_{2 \times 2} \quad (17)$$

where W_c is the camera residual weight. During this uncertainty estimation process, the uncertainty from the IMU measurement itself is fixed and can be neglected numerically. Specifically, for a vehicle moving at 1 m/s, and with a camera working at 10 Hz, the magnitude of $\mathbf{t}_j^i$ in (15) has a typical value of $1 \text{ m/s} \times (1/10 \text{ Hz}) = 0.1 \text{ m}$. Then, for a typical IMU, the white noise density has a typical value of $10^{-3} \text{ (m/s}^2\text{)}(1/\sqrt{\text{Hz}})$. When the IMU is working under 100 Hz, there are 10 IMU measurements that need to be integrated between neighboring camera frames. Therefore, the typical value of the uncertainty caused by IMU integration is about $10 \times (10^{-3} \text{ (m/s}^2\text{)}(1/\sqrt{\text{Hz}}) \times \sqrt{100 \text{ Hz}}) \times (1/100 \text{ Hz})^2 = 10^{-5} \text{ m}$, which is negligible compared with the movement value of 0.1 m.

Here, we also notice that the relative translation between frames $\mathbf{t}_j^i$ in (15) can actually be acquired in two manners. One is the IMU pre-integration terms mentioned earlier, while the other one is the posterior estimation result. The advantages of using IMU pre-integration terms are twofold: first, and most obviously, the IMU pre-integration method has no delay. The uncertainty is calculated simultaneously with the ego-motion estimation. Moreover, the posterior method needs a two-round optimization procedure: one for posterior results and the other for uncertainty estimation, which leads to delay and inefficient computation. Second, using posterior estimation results to calculate uncertainty can lead to a contradiction: the pose estimation relies on depth uncertainty, while depth uncertainty depends on pose estimation. Therefore, as the primary goal of this section is to provide consistent interframe relative uncertainty estimation, we choose the former.

A runtime screenshot of the proposed method is presented in Fig. 5, where the uncertainty increase in turns is demonstrated.

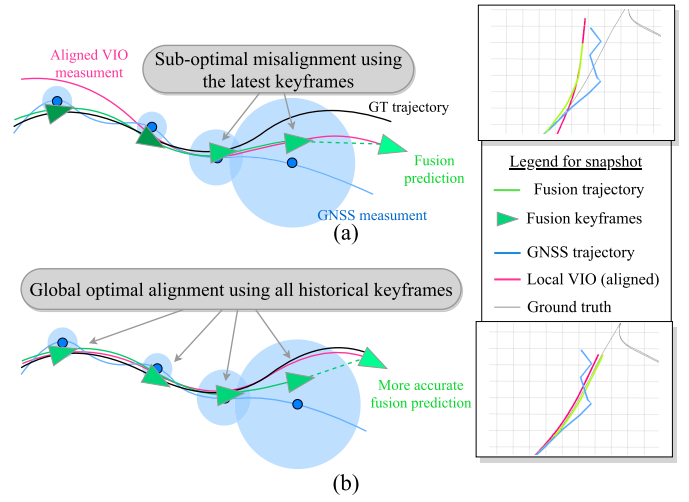


Fig. 6. Comparison between different parameterization strategies under intermittent GNSS-denied environment. Practical runtime snapshots are presented on the right. (a) Without parameterization leads to misalignment and inaccurate fusion prediction. (b) With parameterization of $\mathbf{T}_{\mathcal{M}}^{\mathcal{L}}$ and calculate it explicitly to ensure accurate fusion prediction.

V. 15 DOF GLOBAL AND LOCAL STATES UNDER INTERMITTENT GNSS-DEGRADED ENVIRONMENT

A. Overparameterization on Local-Map-to-LTP Translation

As described in Section II-B, there are four different frames involved when global estimation is expected, and the transformation between them has been shown in (1).

However, in applications requiring global position estimation, $\mathbf{t}_{\mathcal{B}}^{\mathcal{W}}$, the position under the Earth-referenced frame, and $\mathbf{R}_{\mathcal{B}}^{\mathcal{L}}$, the attitude under LTP coordinate system, are the desired states. These two states can be decomposed as

$$\mathbf{R}_{\mathcal{B}}^{\mathcal{L}} = \mathbf{R}_{\mathcal{M}}^{\mathcal{L}} \mathbf{R}_{\mathcal{B}}^{\mathcal{M}} \quad (18)$$

and

$$\mathbf{t}_{\mathcal{B}}^{\mathcal{E}} = \mathbf{R}_{\mathcal{L}}^{\mathcal{E}} (\mathbf{R}_{\mathcal{M}}^{\mathcal{L}} \mathbf{t}_{\mathcal{B}}^{\mathcal{M}} + \mathbf{t}_{\mathcal{M}}^{\mathcal{L}}) + \mathbf{t}_{\mathcal{L}}^{\mathcal{E}}. \quad (19)$$

The decomposed basic translations contain $[\mathbf{R}_{\mathcal{B}}^{\mathcal{M}} | \mathbf{t}_{\mathcal{B}}^{\mathcal{M}}]$, $[\mathbf{R}_{\mathcal{M}}^{\mathcal{L}} | \mathbf{t}_{\mathcal{M}}^{\mathcal{L}}]$, and $[\mathbf{R}_{\mathcal{L}}^{\mathcal{E}} | \mathbf{t}_{\mathcal{L}}^{\mathcal{E}}]$, which have 6, 6, and 3 DoF, respectively. Among these three pairs of transformations, only $[\mathbf{R}_{\mathcal{B}}^{\mathcal{M}} | \mathbf{t}_{\mathcal{B}}^{\mathcal{M}}]$ is motion-related, while others are coordinate-transform-related. In particular, $[\mathbf{R}_{\mathcal{M}}^{\mathcal{L}} | \mathbf{t}_{\mathcal{M}}^{\mathcal{L}}]$ contains information on how the $\mathcal{F}_{\mathcal{M}}$ is aligned with the LTP, i.e., the geodetic orientation and translation. $[\mathbf{R}_{\mathcal{L}}^{\mathcal{E}} | \mathbf{t}_{\mathcal{L}}^{\mathcal{E}}]$ indicates where the LTP is located on Earth. Therefore, $\mathbf{T}_{\mathcal{B}}^{\mathcal{M}} = [\mathbf{R}_{\mathcal{B}}^{\mathcal{M}} | \mathbf{t}_{\mathcal{B}}^{\mathcal{M}}]$ is chosen as the main state in backend optimization.

For the LTP related auxiliary states, we simply used $[\mathbf{R}_{\mathcal{M}}^{\mathcal{L}} | \mathbf{t}_{\mathcal{M}}^{\mathcal{L}}]$ to transfer the states under $\mathcal{F}_{\mathcal{M}}$ into $\mathcal{F}_{\mathcal{L}}$, by

$$\mathbf{R}_{\mathcal{B}}^{\mathcal{L}} = \mathbf{R}_{\mathcal{M}}^{\mathcal{L}} \mathbf{R}_{\mathcal{B}}^{\mathcal{M}} \quad (20)$$

and

$$\mathbf{t}_{\mathcal{B}}^{\mathcal{L}} = \mathbf{R}_{\mathcal{M}}^{\mathcal{L}} \mathbf{t}_{\mathcal{B}}^{\mathcal{M}} + \mathbf{t}_{\mathcal{M}}^{\mathcal{L}}. \quad (21)$$

This contributes a significant feature of our system that, we do not treat $[\mathbf{R}_{\mathcal{M}}^{\mathcal{L}} | \mathbf{t}_{\mathcal{M}}^{\mathcal{L}}]$ as redundancy for $\mathcal{F}_{\mathcal{L}}$, but calculate it explicitly. We have discussed the advantages of

choosing these overparameterization states in the rest of this section and described the detailed optimization formulation in Section VI-A.

Particularly, if $[\mathbf{R}_B^{\mathcal{L}}|\mathbf{t}_B^{\mathcal{L}}]$ is chosen to be the main state, then the measurement for these states would be

$$\mathbf{z}^{\mathcal{L}} = \mathbf{T}_{\mathcal{M}}^{\mathcal{L}} \mathbf{z}^{\mathcal{M}} \quad (22)$$

where $\mathbf{z}^{\mathcal{M}}$ is the VIO estimation result from (5). Here, $[\mathbf{R}_{\mathcal{M}}^{\mathcal{L}}|\mathbf{t}_{\mathcal{M}}^{\mathcal{L}}]$ would be estimated in the optimization formulation (which will be discussed later in Section VI-A). Typically, $\mathbf{T}_{\mathcal{M}}^{\mathcal{L}}$ is estimated by aligning the latest pose under LTP with the latest pose under $\mathcal{F}_{\mathcal{M}}$, as shown in Fig. 6(a). This gives

$$\mathbf{T}_{\mathcal{M}}^{\mathcal{L}} = (\mathbf{z}^{\mathcal{M}} \mathbf{T}_B^{\mathcal{L}-1})^{-1} \quad (23)$$

where $\mathbf{T}_B^{\mathcal{L}}$ is the latest fusion result under LTP (if $\mathbf{T}_B^{\mathcal{L}}$ were chosen to be the main state). Therefore, $[\mathbf{R}_{\mathcal{M}}^{\mathcal{L}}|\mathbf{t}_{\mathcal{M}}^{\mathcal{L}}]$ would become redundancy when GNSS measurements are stable and accurate enough.

However, when GNSS is intermittently degraded, suboptimal estimation using the latest keyframe would present on $\mathbf{T}_B^{\mathcal{L}}$ using (23), as shown in Fig. 6(a). This bias would be further propagated into measurement through (22), which consequentially harms the fusion accuracy by setting biased initial value to the next optimization iteration, and hence, causing suboptimal alignment of the local VIO measurement. This phenomenon is particularly noticeable when the vehicle travels from outdoor to indoor: the alignment uses the latest-but-degraded GNSS measurement only to align the local measurement.

Oppositely, calculating $[\mathbf{R}_{\mathcal{M}}^{\mathcal{L}}|\mathbf{t}_{\mathcal{M}}^{\mathcal{L}}]$ explicitly based on all historical GNSS measurements gives a global optimal alignment of the local map frame $\mathcal{F}_{\mathcal{M}}$ with the geodetic LTP $\mathcal{F}_{\mathcal{L}}$. This optimal alignment guarantees the next initial pose being correctly set, avoiding the optimization falling into a local minimum. From the perspective of data usage, the proposed method does not require extra data but took a better usage of existing data. This process is shown in Fig. 6(b).

B. Global States as Bias Terms

As shown in (19), the proposed method detaches the global information from the motion states and optimizes it explicitly. Furthermore, for $\mathcal{F}_{\mathcal{E}}$ related auxiliary states, we do not directly use $\mathbf{T}_{\mathcal{L}}^{\mathcal{E}}$ to convert the pose from $\mathcal{F}_{\mathcal{L}}$ to $\mathcal{F}_{\mathcal{E}}$. This improves global positioning performance under intermittently GNSS-degraded circumstances. This section illustrates the benefits of this choice.

To elucidate the proposed method and compare it with existing methods, we begin with the frame registration procedure, as shown in Fig. 7. In Fig. 7(a), the local VIO system is initialized at a point O with its coordinate frame $\mathcal{F}_{\mathcal{M}}$. O is located somewhere with no GNSS signal, or the signal quality is under the acceptable threshold. For example, in our case, this threshold is set to 50-m horizontal dilution of precision (HDOP). Therefore, no LTP can be determined at O . Then, at time t_a , point A , the first acceptable GNSS measurement is received, and an LTP coordinate $\mathcal{F}_{\mathcal{L}}$ is accordingly created.

Next, at time t_b , the vehicle moves to point B , where a relatively more accurate GNSS measurement (also known as with a smaller HDOP) is received. Then, the system generally has two-side strategy choices.

- 1) *Strategy I*: Re-initializing a new LTP coordinate. As shown in Fig. 7(b), at time t_b , point B , a better GNSS signal is received; therefore, a new LTP is re-initialized based on the new-coming GNSS.
- 2) *Strategy II (Our Choice)*: Optimizing on LTP bias. As proposed in Fig. 7(d), the original LTP is reserved. Instead of optimizing the origin, the bias of the trajectory with respect to the origin is optimized.

Among these three choices, the strategy I is the most naive one. It can be easily misguided by an overconfident measurement under intermittently GNSS-degraded circumstances.

In comparison, strategy II adds a bias term $\mathbf{t}[\text{bias}]$ inside the LTP; therefore, (19) becomes

$$\mathbf{t}_B^{\mathcal{E}} = \mathbf{R}_{\mathcal{L}}^{\mathcal{E}} (\mathbf{R}_{\mathcal{M}}^{\mathcal{L}} \mathbf{t}_B^{\mathcal{M}} + \mathbf{t}_{\mathcal{M}}^{\mathcal{L}} + \mathbf{t}^{\text{bias}}) + \mathbf{t}_{\mathcal{L}}^{\mathcal{E}}. \quad (24)$$

Then, strategy II optimizes on $\mathbf{t}^{\text{bias}}$. By doing so, this strategy benefits in two aspects: first, it is a multi-observation-based optimization strategy, and thus, improves the accuracy and robustness of the origin. Second, by fixing $\mathbf{R}_{\mathcal{L}}^{\mathcal{E}}$ and $\mathbf{t}_{\mathcal{L}}^{\mathcal{E}}$ to constants, this strategy relieves the computation demand by avoiding calculating $\mathbf{R}_{\mathcal{L}}^{\mathcal{E}}$ and its derivatives repeatedly. Specifically, as mentioned in Section II-B, $\mathbf{R}_{\mathcal{L}}^{\mathcal{E}}$ is a nonlinear function of $\mathbf{t}_{\mathcal{L}}^{\mathcal{E}}$ and has a complicated closed form. If we choose to optimize the origin of the LTP directly, which is $\mathbf{t}_{\mathcal{L}}^{\mathcal{E}}$ in (19), each optimization iteration needs to recompose $\mathbf{R}_{\mathcal{L}}^{\mathcal{E}}$, which increases the computation consumption.

Therefore, the $\mathcal{F}_{\mathcal{E}}$ related auxiliary states are represented by $\mathbf{t}^{\text{bias}}$. Then, the full states considered in the proposed backend on frame k is

$$\mathbf{x}_k^{\mathcal{M}} = [\mathbf{q}_{B_k}^{\mathcal{M}^T} \quad \mathbf{t}_{B_k}^{\mathcal{M}^T}]^T \quad (25)$$

with a unique global auxiliary state

$$\mathbf{x}^{\text{global}} = [\mathbf{q}_{\mathcal{M}}^{\mathcal{L}^T} \quad \mathbf{t}_{\mathcal{M}}^{\mathcal{L}^T} \quad \mathbf{t}^{\text{bias}^T}]^T. \quad (26)$$

These states for each step have an overall DoF of 15, including 6 local DoF, 6 local-to-LTP coordinate transformation DoF, and 3 global DoF. This decomposition-based state registration is summarized in Algorithm 1.

VI. DUAL-STAGE ALTERNATING GLOBAL OPTIMIZATION

A. Global Pose-Graph Formulation

On the backend fusion, we adopted the mature pose-graph framework [7], [14], [28] for multisensor fusion, but resorted to attain Earth-referenced global results instead of LTP results.

The states of the global optimization problem are defined as

$$\mathcal{X}^{\mathcal{E}} = [\mathbf{x}^{\text{global}^T} \quad \mathbf{x}_0^{\mathcal{M}^T} \quad \mathbf{x}_1^{\mathcal{M}^T} \quad \dots \quad \mathbf{x}_n^{\mathcal{M}^T}]^T \quad (27)$$

where $\mathbf{x}^{\text{global}}$ and $\mathbf{x}_k^{\mathcal{M}}$ are defined in (26) and (25). n is the total number of frames.

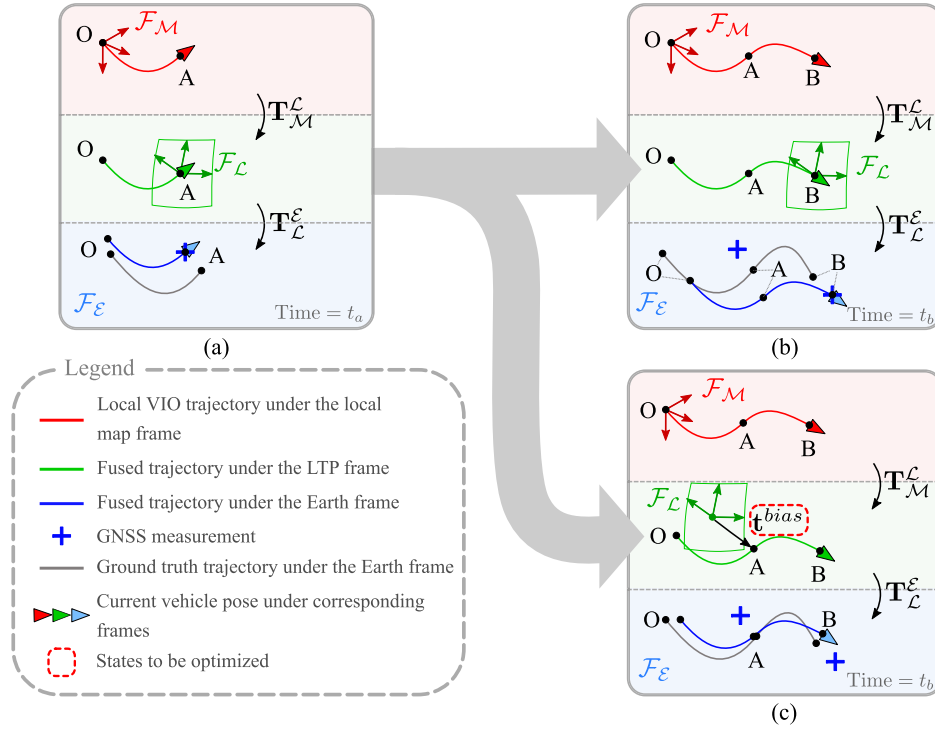


Fig. 7. LTP initialization under intermittent GNSS-denied environment. (a) LTP initialization when the first GNSS solution is received. (b) Strategy I: LTP re-initialization when a better GNSS solution is received. (c) Strategy II (our choice): optimize LTP with bias term when a better GNSS solution is received.

Algorithm 1 Pseudocode for States Registration

```

begin
  VIO initialization under  $\mathcal{F}_M$ ;
   $\mathbf{q}_B^M = 1$ ;
   $\mathbf{t}_B^M = 0$ ;
  while sequence not end do
    Estimate  $\mathbf{q}_B^M$  and  $\mathbf{t}_B^M$ ;
    if GNSS quality presents within threshold then
      if LTP not initialized then
        Initialize LTP  $\mathcal{F}_L$ ;
         $\mathbf{t}^{bias} = 0$ ;
         $\mathbf{q}_M^L = 1$ ;
         $\mathbf{t}_M^L = 0$ ;
      else
        Compose global state initial value:
         $\mathbf{R}_L^E(\mathbf{R}_M^L \mathbf{t}_B^M + \mathbf{t}_M^L + \mathbf{t}^{bias}) + \mathbf{t}_L^E$ ;
      end
      Optimization (Sec. VI) of  $\mathbf{q}_B^M, \mathbf{t}_B^M, \mathbf{q}_M^L, \mathbf{t}_M^L$ 
      and  $\mathbf{t}^{bias}$ ;
    end
  end

```

The backend fusion optimization can be summarized as an maximum likelihood estimation (MLE) problem as

$$\mathcal{X}^{\mathcal{E}*} = \arg \min_{\mathcal{X}^{\mathcal{E}}} \sum_{t=0}^n (\|\mathbf{z}_t^M - h_t^M(\mathcal{X}^{\mathcal{E}})\|_{\Sigma_t^M}^2 + \|\mathbf{z}_t^E - h_t^E(\mathcal{X}^{\mathcal{E}})\|_{\Sigma_t^E}^2). \quad (28)$$

In (28), $\|\mathbf{z}_t^M - h_t^M(\mathcal{X}^{\mathcal{E}})\|_{\Sigma_t^M}^2 \triangleq \mathbf{r}_t^M$ is the local residual, where $\mathbf{z}_t^M$ is the local relevant measurement with covariance Σ_t^M . $\|\mathbf{z}_t^E - h_t^E(\mathcal{X}^{\mathcal{E}})\|_{\Sigma_t^E}^2 \triangleq \mathbf{r}_t^E$ is the global residual, where the global measurement $\mathbf{z}_t^E$ has the covariance of Σ_t^E .

For the local residual $\mathbf{r}_t^M$, a frame-differential relative VIO estimation is used as the measurement

$$\begin{aligned} \mathbf{r}_t^M &= \|\mathbf{z}_t^M - h_t^M(\mathbf{x}_{t-1}^L, \mathbf{x}_t^L)\|_{\Sigma_t^M}^2 \\ &= \left\| \begin{bmatrix} \mathbf{R}_{t-1}^{VIO^{-1}}(\mathbf{t}_{t-1}^{VIO} - \mathbf{t}_t^{VIO}) \\ q\{\mathbf{R}_{t-1}^{VIO^{-1}} \mathbf{R}_t^{VIO}\} \end{bmatrix} \right\|_{\Sigma_t^M}^2 \\ &\triangleq \left\| \begin{bmatrix} \mathbf{R}_{B_{t-1}}^{M^{-1}}(\mathbf{t}_{B_{t-1}}^M - \mathbf{t}_{B_t}^M) \\ q\{\mathbf{R}_{B_{t-1}}^{M^{-1}} \mathbf{R}_{B_t}^M\} \end{bmatrix} \right\|_{\Sigma_t^M}^2 \end{aligned} \quad (29)$$

where $(\mathbf{R}_t^{VIO}, \mathbf{t}_t^{VIO})$ and $(\mathbf{R}_{t-1}^{VIO}, \mathbf{t}_{t-1}^{VIO})$ are the local pose estimation by VIO as mentioned in (2).

Moreover, the GNSS position under the ECEF frame is

$$\mathbf{z}_t^E = C(\mathbf{p}_t). \quad (30)$$

Therefore, the global residual is

$$\begin{aligned} \mathbf{r}_t^E &= \|\mathbf{z}_t^E - h_t^E(\mathbf{x}^{\text{global}}, \mathbf{x}_t^L)\|_{\Sigma_t^E}^2 \\ &= \|C(\mathbf{p}_t) - (q_L^E(q_M^L \mathbf{t}_{B_t}^M + \mathbf{t}_t^L + \mathbf{t}^{bias}) + \mathbf{t}_t^E)\|_{\Sigma_t^E}^2. \end{aligned} \quad (31)$$

The corresponding pose graph structure is illustrated as a factor graph [29] in Fig. 8. In our implementation, this pose graph is optimized every time a GNSS signal is received with the Ceres solver [30] toolset.

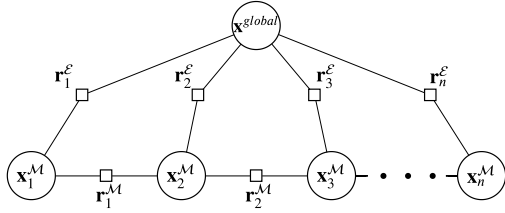


Fig. 8. Factor graph model for the MLE problem (28). The circles represent the states, while the squares represent the factors.

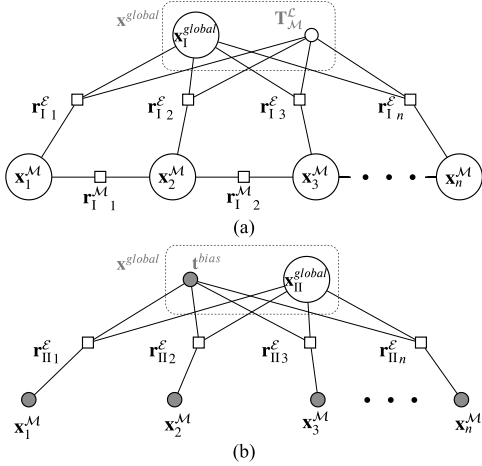


Fig. 9. Dual-stage alternating optimization factor graph model, where (a) for stage I and (b) for stage II. The circles represent the state variables, while the squares represent the factors. The solid circles are constants within the factors, and we pick them out to illustrate the structure.

B. Dual-Stage Alternating Optimization

The proposed optimization problem (28) in Section VI-A presents two peculiarities: first, the state variable $\mathbf{q}_{\mathcal{M}}^{\mathcal{L}}$ is coupled with $\mathbf{t}_{B_i}^{\mathcal{M}}$. Second, under the GNSS intermittent degraded environment, the observation significance of the global measurement is much lower than that of the local relative measurement.

Hence, we employed an AM algorithm to efficiently solve the optimization problem (28). The main idea of the AM algorithm is to solve an intractable global optimization problem by successively performing minimization on split subblocks of variables when there is a natural division of variables [31].

In general, for an optimization problem over two sets of variables

$$\min_{\mathbf{x} \in \mathbb{R}^n, \mathbf{y} \in \mathbb{R}^m} f(\mathbf{x}, \mathbf{y}) \quad (32)$$

where $f: \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}$ is a continuous function. Typically, an AM algorithm solves (32) by alternatively updating the two variable blocks. In our particular problem, the coupled states [(27)] can be naturally split into two groups. The first group includes the motion-related states, including all local pose $\mathbf{x}_k^{\mathcal{M}}$ and global position $\mathbf{t}^{\text{bias}}$. The second group includes the coordinates-alignment related states, $\mathbf{q}_{\mathcal{M}}^{\mathcal{L}}$ and $\mathbf{t}_{\mathcal{M}}^{\mathcal{L}}$. Therefore, we split the optimization into two stages based on this grouping.

Stage I was designed for optimizing the local poses and calculating the LTP origin. The graph model is shown in Fig. 9(a). We rewrote the variable states in (27) as

$$\mathcal{X}_I^{\mathcal{E}} = [\mathbf{x}_1^{\text{global}^T} \mathbf{x}_0^{\mathcal{M}^T} \mathbf{x}_1^{\mathcal{M}^T} \dots \mathbf{x}_n^{\mathcal{M}^T}]^T \quad (33)$$

where

$$\mathbf{x}_1^{\text{global}} = \mathbf{t}^{\text{bias}}. \quad (34)$$

The objective in (28) remains the same and so do the residuals $\mathbf{r}_{l_i}^{\mathcal{M}}$ and $\mathbf{r}_{l_i}^{\mathcal{E}}$. The only difference is the LTP-related transform $\mathbf{q}_{\mathcal{M}}^{\mathcal{L}}$ and $\mathbf{t}_{\mathcal{M}}^{\mathcal{L}}$ in global residual $\mathbf{r}_{l_i}^{\mathcal{E}}$ are fixed.

Then, stage II is in charge of optimizing LTP direction and translation only. The LTP-related transform $\mathbf{q}_{\mathcal{M}}^{\mathcal{L}}$ and $\mathbf{t}_{\mathcal{M}}^{\mathcal{L}}$ are calculated, as shown in Fig. 9(b). The variable states in this stage reduce to

$$\mathcal{X}_{II}^{\mathcal{E}} = \mathbf{x}_{II}^{\text{global}} = [\mathbf{q}_{\mathcal{M}}^{\mathcal{L}^T} \mathbf{t}_{\mathcal{M}}^{\mathcal{L}^T}]^T \quad (35)$$

and the MLE problem becomes

$$\mathcal{X}_{II}^{\mathcal{E}*} = \arg \min_{\mathcal{X}_{II}^{\mathcal{E}}} \sum_{t=0}^n \|\mathbf{z}_t^{\mathcal{E}} - h_{\text{II}t}^{\mathcal{E}}(\mathcal{X}_{II}^{\mathcal{E}})\|_{\Sigma_t^{\mathcal{E}}}^2 \quad (36)$$

where $\|\mathbf{z}_t^{\mathcal{E}} - h_{\text{II}t}^{\mathcal{E}}(\mathcal{X}_{II}^{\mathcal{E}})\|_{\Sigma_t^{\mathcal{E}}}^2 \triangleq \mathbf{r}_{\text{II}t}^{\mathcal{E}}$ has the same form as (31). These two stages are optimized in an alternating manner and updated every time a new GNSS solution is received.

VII. IMPLEMENTATIONS AND EXPERIMENTAL RESULTS

The whole system was implemented under Ubuntu 1604, and Ceres Solver [30] toolset was used in the graph optimization. The system modules were connected using the Robot Operating System [32].

We performed two sets of experiments to evaluate the proposed GVINS system. The first set of experiments compared our system with state-of-the-art methods on public data sets with some simulated modification. Then, in the second set of real-world experiments, we tested the proposed system under indoor-outdoor switching scenarios to evaluate the system under GNSS intermittent-denied environments.

A. Simulated Data Set Comparison

We first test our system on KITTI [16] raw-data data set. This data set was collected from a ground vehicle equipped with a stereo camera, 3-D Lidar, and GNSS/IMU.

To simulate intermittent GNSS-degraded scenarios, the data set was used with some simulated modifications.

- 1) *Stereo Images*: We used the stereo images as it is, with a rate of 10 Hz and sizes of 1241×376 .
- 2) *IMU Data*: IMU data were interpolated and resynchronized at 100 Hz to cooperate with the visual images.
- 3) *GNSS Data*: The original KITTI's GNSS data were collected in residential areas with open sky. It has high accuracy (~ 0.1 m), and therefore, being treated as ground truth. To simulate intermittent GNSS-degraded scenarios, we artificially added noise into GNSS measurements in some parts of the trajectory.

The reprocessing procedure is detailed in the Appendix.

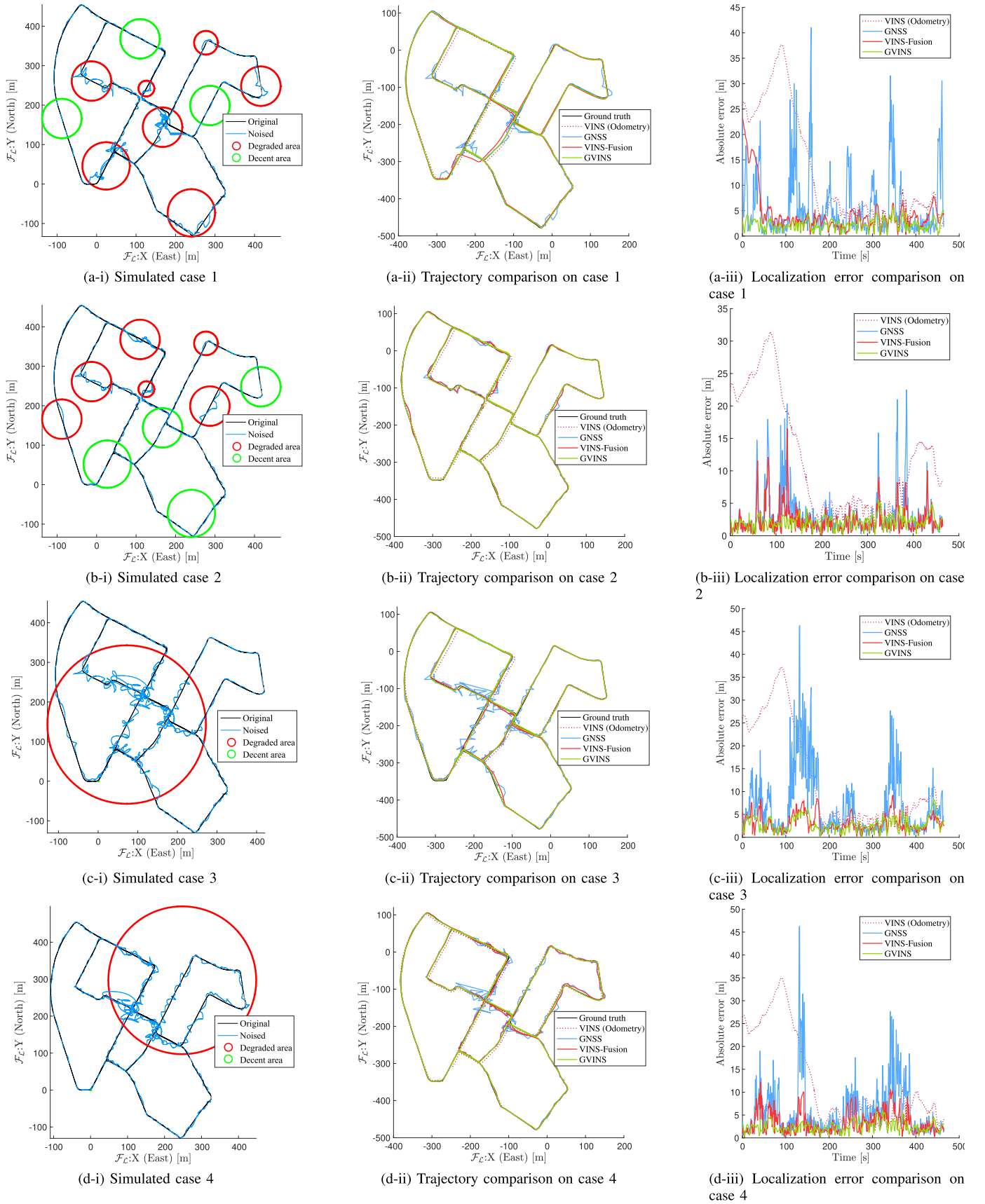


Fig. 10. Simulated intermittent GNSS-denied environments test cases and result comparisons. (a-i) Simulated case 1. (a-ii) Trajectory comparison on case 1. (a-iii) Localization error comparison on case 1. (b-i) Simulated case 2. (b-ii) Trajectory comparison on case 2. (b-iii) Localization error comparison on case 2. (c-i) Simulated case 3. (c-ii) Trajectory comparison on case 3. (c-iii) Localization error comparison on case 3. (d-i) Simulated case 4. (d-ii) Trajectory comparison on case 4. (d-iii) Localization error comparison on case 4.

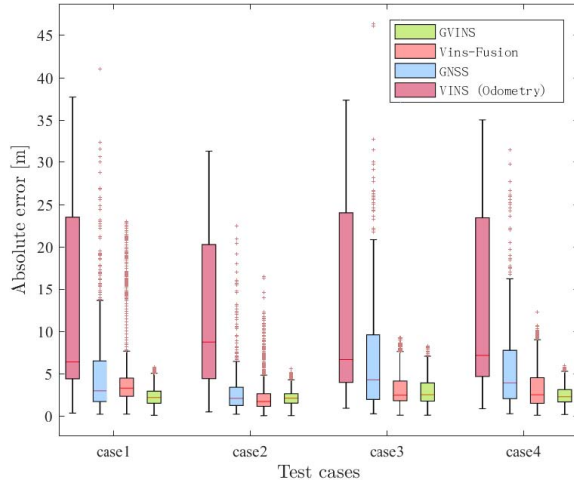


Fig. 11. Statistical error comparison on simulated intermittent GNSS-degraded environments.

We chose the *2011_10_03_drive_0027* sequence from the KITTI raw-data data set as the base data, as it is the most commonly used and most challenging sequence in the data set. Then, we consider four different noise-strategies, simulating different practical cases. These four cases are shown in the first column of Fig. 10, including in the following.

- 1) *Case 1 [Fig. 10(a-i)]*: Intermittent GNSS-degraded areas scattering over the map. The vehicle starts from a GNSS-degraded area.
- 2) *Case 2 [Fig. 10(b-i)]*: A similar scenario like case 1, yet with GNSS-degraded areas re-arranged and the vehicle starts from a GNSS-decent area.
- 3) *Case 3 [Fig. 10(c-i)]*: One large, uniformly distributed GNSS-degraded area. The vehicle starts from this GNSS-degraded area.
- 4) *Case 4 [Fig. 10(d-i)]*: A similar scenario like case 3 but the vehicle starts out-of the GNSS-degraded area.

The noise has three different levels: inside the red circle, GNSS signals are considered as degraded, and the errors are magnified 50 to 200 times. This simulates areas of the urban canyon or indoor scenarios.

In contrast, the green circles represent GNSS-decent areas, where the GNSS errors are magnified with a factor of 5, reaching a standard deviation (STD) order of 0.5 m. This simulates areas of open-sky scenarios. Finally, in all the rest areas, the errors are magnified by a factor of 10, resulting in an accuracy of around 1 m. This simulates the ordinary positioning accuracy of a low-cost GNSS receiver working in an urban area.

We compared the proposed method, GVINS, with the state-of-the-art fusion method, VINS-Fusion [7]. The trajectory results and the absolute errors are plotted in the second and third columns of Fig. 10. To compare the localization results under a unified stander, we transformed all the trajectories into the same LTP frame and calculated localization errors using the same metric. We also provided the results of VINS [5] running under odometry mode and GNSS solutions as references. The results of VINS are aligned to the LTP frame

TABLE I
NUMERIC ABSOLUTE ERROR COMPARISON OF THE LOCALIZATION RESULTS ON SIMULATED INTERMITTENT GNSS-DEGRADED ENVIRONMENTS

Cases	Statistics	VINS (Odometry)	GNSS	VINS-Fusion	GVINS
Baseline (no noise)	MSE(m)	12.59	-	0.21	0.21
	STD(m)	10.88	-	0.13	0.13
	Max Error(m)	36.84	-	1.02	1.02
Case 1	MSE(m)	12.54	5.56	4.41	2.26
	STD(m)	11.01	6.29	4.09	1.03
	Max Error(m)	37.68	41.01	23.00	5.73
Case 2	MSE(m)	12.04	3.19	2.31	2.13
	STD(m)	8.94	3.44	2.00	0.89
	Max Error(m)	31.33	22.47	16.50	5.56
Case 3	MSE(m)	12.60	6.87	3.09	2.86
	STD(m)	11.12	7.09	1.85	1.51
	Max Error(m)	37.32	46.33	9.20	8.25
Case 4	MSE(m)	12.50	5.93	3.30	2.46
	STD(m)	10.20	6.02	2.33	1.09
	Max Error(m)	35.00	46.33	12.20	5.94

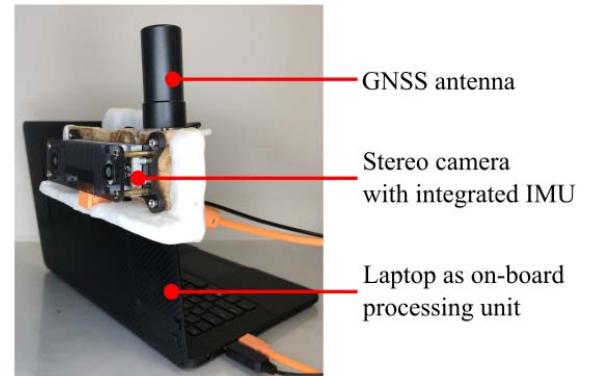


Fig. 12. Hand-held testbed mounted to a laptop.

so that the absolute error can be evaluated. The statistical result comparison is provided in Fig. 11 and Table I. Table I also provides a baseline result that no additional noise is added on the GNSS data for references.

Clearly, the results have shown a significant accuracy boost of GVINS over the compared methods in all four cases. GVINS has the lowest mean square error and STD. Meanwhile, GVINS gives the best performance in terms of the max error and the number of outliers. When the GNSS quality is decent, the proposed method degenerates into VINS-Fusion and performs identically.

It can be further noticed that in cases 1 and 2, where the GNSS is more volatile, the proposed method attains a

more significant improvement. This is benefited from the overparameterization method in Section V and together with the followed optimization formulation. All these factors make the system more insensitive to the perturbation of GNSS. Moreover, in cases that start from GNSS-degraded areas, adjusting LTP by origin bias results in higher accuracy. This is shown in cases 1 and 3, e.g., near the starting point around $(-280, -300)$ in Fig. 10(a-ii) and (c-ii).

On the computation efficiency side, the proposed method consumes an extra runtime of about 19.37% on average compared with the traditional method. This increment is under expectation, as both the node number and DoF of the problem increase. However, we argued that it is worthy to attain an accuracy boost up to 50% with a 20% extra computation time cost.

B. Real-World Experiments

To evaluate the proposed system under real-world applications, we first set up a hand-hold testbed, as shown in Fig. 12. Simplified from our previous setup [33], this testbed contains a Mynteye S2110-95 stereo camera with built-in IMU of Bosch BMI088, and a u-blox F9P GNSS receiver. We sampled vision data under VGA resolution at 20 Hz with an IMU of 200 Hz, and the GNSS report rate is 1 HZ. The system was tested on-board using a laptop with an Intel Core-i7 CPU.

The test environment is an office park, as shown in Fig. 13. This environment contains three typical scenes, including the GNSS-denied indoor scene, GNSS-degraded obstructed scene and GNSS-decent open-sky scene. To provide a numerical evaluation of the tests, we further set up 15 control points by averaging the measurements of a real time kinematics differential GNSS base station over 10 min at each point. By designating trajectories through the control points, we tested the system's adaptation under various GNSS status.

The test trajectories are as follows.

- 1) *Trajectory 1*: As shown in Fig. 14(a-i), this trajectory contains two loops. The loops start from GNSS-denied indoors, go through a near-building scene to an open-sky scene, and then enter into indoors again. The GNSS localization accuracy varies dramatically in the trajectory. The total trajectory length is about 560 m.
- 2) *Trajectory 2*: As shown in Fig. 14(b-i), this trajectory takes a full loop around the building. The GNSS localization accuracy varies little in the trajectory. The total trajectory length is about 330 m.

The resulting trajectory comparison is shown in Fig. 14(a-ii) and (b-ii), with the error comparison on control points in Fig. 14(a-iii) and (b-iii). The statistical error comparison on control points is shown in Table II.

It can be seen from Table II that the aligned pure odometry has a decent accuracy, which is as expected since the environment is rich in textures; meanwhile, the lighting condition is adequate. However, the quality variation of GNSS causes an accuracy drop on VINS-Fusion, due to its underlying overconfidence on GNSS measurements and misalignment of the local and global measurements. We zoomed near $(-60, 5)$ of Fig. 14(a-ii) in Fig. 15 to demonstrate this advantage. This



Fig. 13. Experiment scenarios containing indoor, obstructed, and open-sky areas. Control points are also shown in white dots.

TABLE II
NUMERIC ABSOLUTE ERROR COMPARISON ON CONTROL POINTS OF THE LOCALIZATION RESULTS IN REAL-WORLD TESTS

Cases	Statistics	VINS (Odometry)	GNSS	VINS- Fusion	GVINS
Trajectory 1	MSE(m)	3.45	3.99	3.98	2.72
	STD(m)	2.19	3.79	4.25	2.12
	Max Error(m)	8.46	16.25	15.34	8.06
Trajectory 2	MSE(m)	3.58	3.61	3.99	2.76
	STD(m)	2.05	1.39	1.45	1.43
	Max Error(m)	8.51	5.97	6.43	4.34

place is near the doorway of the building; therefore, GNSS undergoes a dramatic accuracy variation. The misalignment of the local and global measurements in VINS-fusion harms the accuracy of the followed indoor scenario path (the indoor path should be overlapped).

Furthermore, as trajectory 1 is initialized indoors, the GNSS is unreliable in the beginning. With the help of the LTP bias term mentioned in Section V-B, the proposed system achieves a significant improvement in the beginning stage.

VIII. CONCLUSION

In this article, we proposed a global positioning system adept at localization in intermittent GNSS-degraded

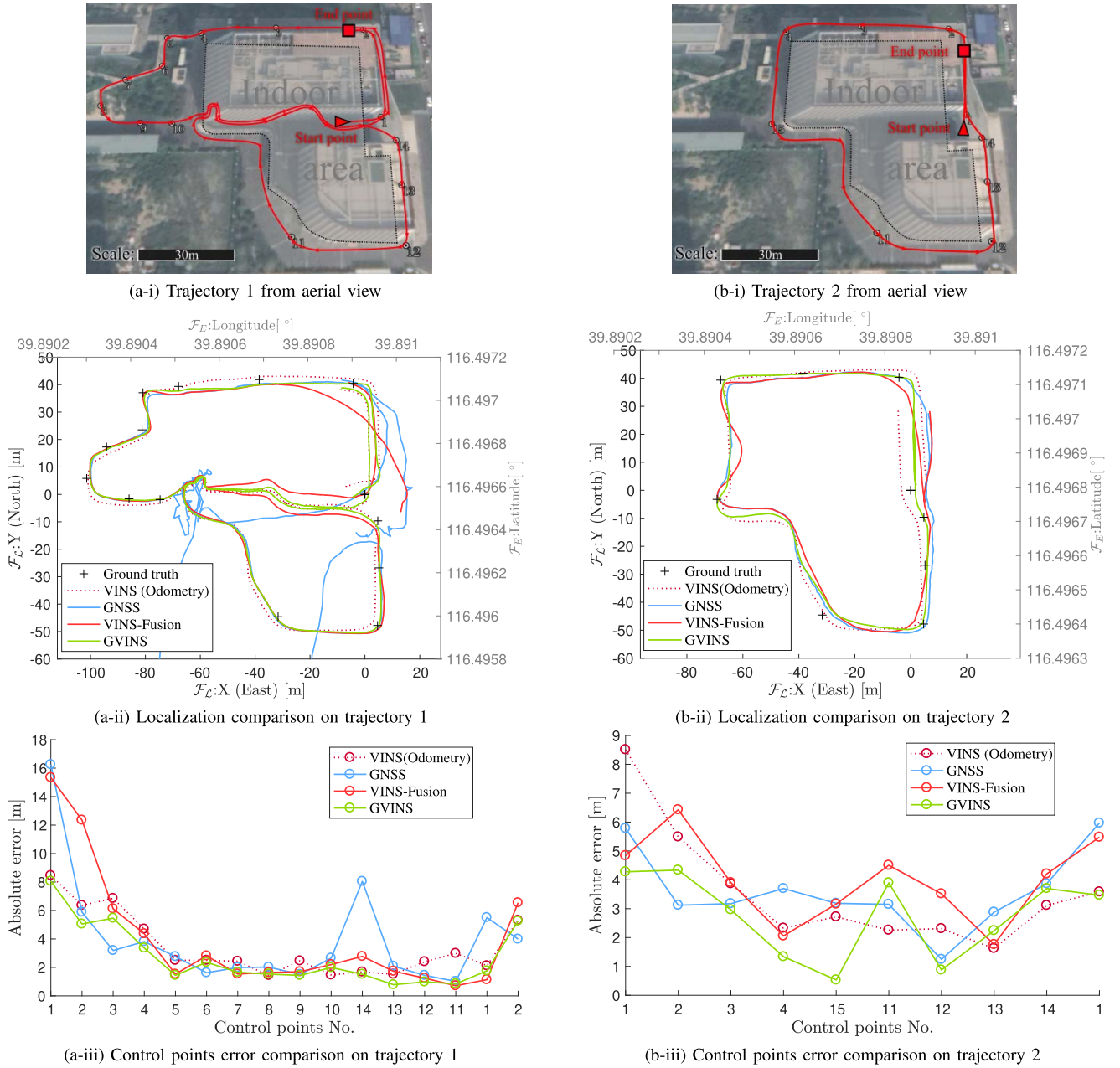


Fig. 14. Real-world tests and result comparisons. (a-i) Trajectory 1 from aerial view. (a-ii) Localization comparison on trajectory 1. (a-iii) Control points error comparison on trajectory 1. (b-i) Trajectory 2 from aerial view. (b-ii) Localization comparison on trajectory 2. (b-iii) Control points error comparison on trajectory 2.

environments by adaptively fusing GNSS, vision, and IMU measurements. First, an IMU pre-integration based ego-motion uncertainty estimation method was proposed to obtain proper weights of fusion sources. Then, we innovatively introduced three extra terms, with a total of nine extra DoF into the fusion optimization formulation, which made the system more stable and accurate in intermittent GNSS-degraded environments. These terms include a local-map-to-LTP pose and LTP origin bias. Furthermore, to overcome the difficulty of high-DoF optimization, we split the optimization procedure into two stages. Finally, our system was evaluated on both the public data set and real-world tests in comparison with state-of-the-art. The results demonstrated the higher accuracy and better

robustness of the proposed system in intermittent GNSS-degraded environments.

APPENDIX INTERPOLATION, RESYNCHRONIZATION, AND NOISE SIMULATION ON KITTIT RAW-DATA DATA SET

There are two different subsets provided by the KITTI raw-data data set, namely, the *unsynced + unrectified* data and the *synced + rectified* data. In *synced + rectified* data, the images have been rectified and undistorted. However, the IMU data in *synced + rectified* data have also been downsampled to match the image's rate, which is 10 Hz. This is too low to

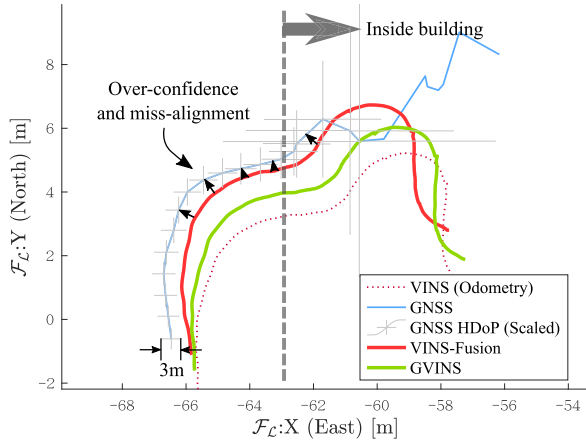


Fig. 15. Zoom around $(-60, 5)$ in Fig. 14(a-ii), with GNSS HDOP under 50-m overlapped. For readability, historical trajectories are hidden and HDOP is scaled by 0.1.

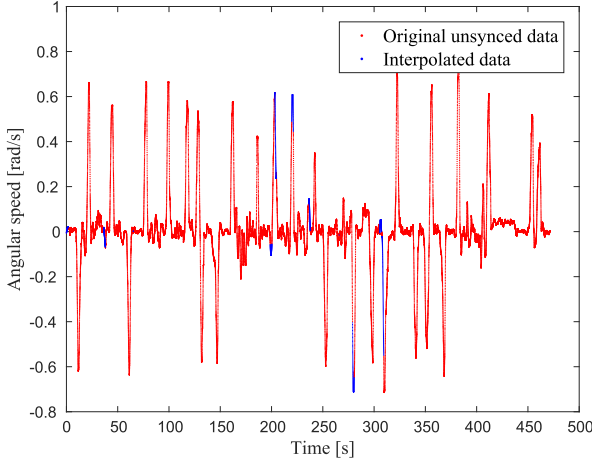


Fig. 16. KITTI *unsynced* + *unrectified* raw-data interpolation.

tightly couple IMU data with vision. On the other hand, in the *unsynced* + *unrectified* data, the IMU data remains 100-Hz output rate, yet has interruption and delay with the visual images (as shown in Fig. 16). Therefore, to make better use of IMU, we reprocessed the data set via the following steps.

- 1) Using stereo images, with its timestamps from the *synced* + *rectified* data set.
- 2) Using IMU data from the *unsynced* + *unrectified* data set. Interpolating the deficiency and then aligning it with the visual timestamps.
- 3) Using GNSS data from *unsynced* + *unrectified* data set. Interpolating the deficiency and then artificially setting interfered areas by adding noise to simulate GNSS-degraded areas.

We took the set *2011_10_03_drive_0027*, i.e., the corresponding raw data of the “KITTI00” sequence, as an example.

The angular velocity on z -axis ω_z in the *unsynced* data is shown in Fig. 16 by red dots. We used a cubic spline to interpolate the missing data shown by blue dots. Then, we aligned timestamps of the high-frequency *unsynced* data with the *synced* ones. Since there was only a fixed, global

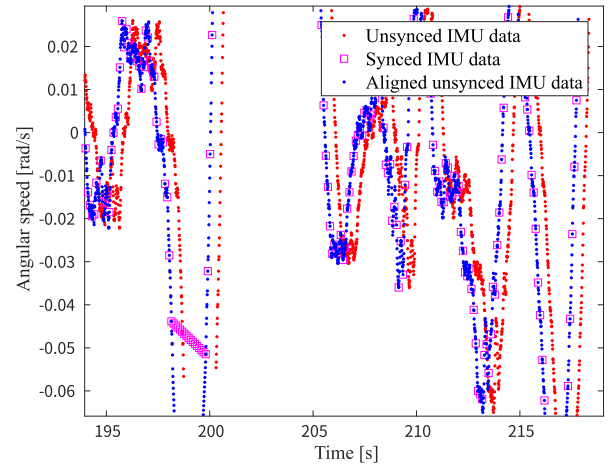


Fig. 17. KITTI *unsynced* + *unrectified* raw-data time alignment.

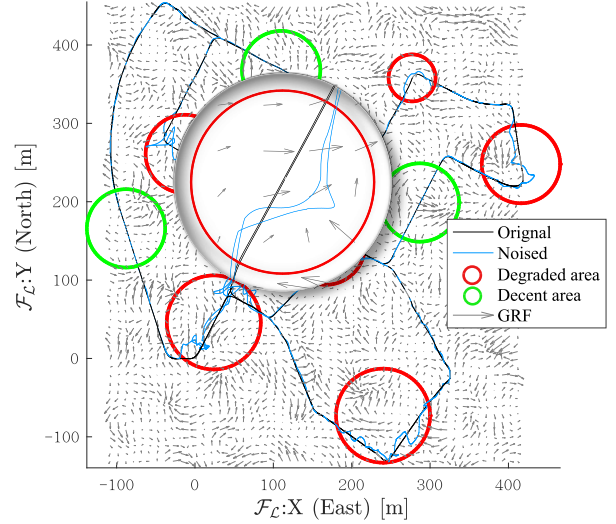


Fig. 18. Noised GNSS trajectory with temporal GM process and GRF as repeatable territory-related multipath bias. The center is enlarged to show this repeatable feature.

constant bias on the timestamp, we simply calculated this bias by an MLE. An alignment result is partially shown in Fig. 17. The nonoverlapped data near 200 s is caused by the difference in interpolation: KITTI *synced* data interpolates the missing data linearly.

Then, for the GNSS data, interpolation and time alignment are also performed similar to IMU. In addition, we downsampled it to 1 Hz and added noise into arbitrarily selected areas as introduced in Section VII-A, to simulate the low-cost GNSS measurements in intermittent GNSS-degraded environments. We simulated this noise as the composition of two components.

- 1) A temporal term represents the errors, including atmosphere error, clock error, earth rotation error, and filter error.
- 2) A spatial term represents the environment-dependent multipath bias.

According to [34], we simplified the temporal term as a Gaussian–Markov (GM) process. Then, for the spatial

multipath bias term, it is principally environment related and requires a dedicated model to simulate. To simplify the model while retaining the feature of environmental correlation and repeatability, we followed the idea proposed in [35] and [36]. Specifically, we created a stationary Gaussian random field (GRF) over the entire terrain using the methods in [37] and [38]. This GRF gives the bias a repeatable behavior depending on territory, as shown in Fig. 18. Finally, we performed an additional cubic spline interpolation on the junction edge to smooth the jumps.

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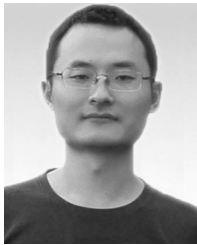
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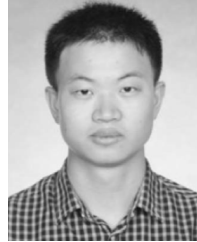
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